

Sandeep Kumar Amgothu

Data Engineering | Machine Learning & AI Safety | Cloud & MLOps | Analytics & Power Platform

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PROFESSIONAL SUMMARY

Software and Data Engineer / Machine Learning Engineer with 5+ years building production-grade AI systems, cloud-native data platforms, and enterprise analytics. Specializes in LLM security and automated red teaming, computer vision, scalable ETL/ELT, and MLOps. Deployed data/ML workloads processing 15M+ records per month with 40% latency reduction; orchestrated 3,000+ adversarial LLM evaluations with 20–50 percentage-point risk reduction; published researcher (ICUAS 2025). Bridges academic innovation and enterprise-grade engineering across AWS, Azure, Spark, Power BI, and modern GenAI stacks.

TECHNICAL SKILLS

Languages & Core: Python, SQL, T-SQL, PySpark, Bash, Java, C++, JavaScript, Scala

Machine Learning & AI: Supervised/unsupervised learning, classification, anomaly detection, PyTorch, TensorFlow, Scikit-learn, XGBoost, feature engineering, hyperparameter tuning, explainable AI

LLM & Generative AI: GPT-4/4o, LLaMA-2/3, BERT, T5, instruction-tuning, LoRA/QLoRA, prompt engineering, few-shot learning, LLM-as-a-judge, LangChain, LlamaIndex, Hugging Face Transformers, hallucination mitigation

AI Safety & Security: Automated red teaming, jailbreak attack generation, PYRIT Crescendo, taxonomy-driven evaluation, safety fine-tuning, LlamaGuard, Detoxify, DeBERTa reward models, RoBERTa hate-speech classifiers

MLOps & Infrastructure: MLflow, DVC, Weights & Biases, experiment tracking, Docker, Kubernetes, CI/CD (GitHub Actions), model monitoring, A/B testing, reproducible pipelines, model lifecycle management

Data Engineering & Big Data: ETL/ELT, Apache Spark, Airflow, Kafka, data lakes, schema design, data quality, incremental loads, dimensional modeling (star/snowflake), AWS (S3, Glue, Lambda, Redshift, EMR), Snowflake, Azure data patterns

Vector Search & Retrieval: FAISS, Pinecone, Milvus, Chroma, RAG pipelines, dense retrieval, semantic search, Elasticsearch, embedding pipelines

Cloud & DevOps: AWS, Azure (Administrator Certified), GCP, Terraform, Linux, REST APIs, FastAPI, Flask, serverless architectures, cost/performance optimization

Databases: PostgreSQL, MySQL, SQL Server, Redshift, MongoDB, Dataverse patterns

Analytics & Visualization: Power BI (datasets, dataflows, DAX, RLS, Import/DirectQuery), Tableau, SQL Server Reporting, Matplotlib, Seaborn, Plotly, Streamlit, dashboard design, KPI definition

Power Platform: Power BI, Power Automate, Power Apps, SharePoint integration

Data Collection & Formats: Selenium, BeautifulSoup, web scraping, NLP document parsing, JSON, CSV, Parquet, REST/JSON APIs

PROFESSIONAL EXPERIENCE

Data Pioneer (Non-Profit)

Remote

Data Analytics & AI Volunteer Engineer | Aug 2025–Apr 30, 2026

- Engineered automated data-quality and anomaly-detection pipelines in Python/SQL for nonprofit program datasets, improving reporting accuracy by 32% and cutting manual review time by 40%.
- Built and automated Power BI/Tableau dashboards tracking volunteer engagement and program KPIs across 5+ initiatives, enabling data-driven decisions for 15+ stakeholders.
- Migrated legacy spreadsheet workflows to cloud-based (AWS/Azure) data pipelines, reducing processing turnaround from days to hours and saving an estimated \$8K annually in operational overhead.
- Architected Power BI semantic models and dashboards on SQL/Dataverse-backed datasets for program KPIs, cutting manual reporting effort by 45% across 3 teams.
- Engineered Python/SQL ETL and nightly ELT-style ingestion for multi-source operational data (JSON/CSV), improving data freshness to under 24 hours.
- Designed dimensional-style schemas (12 fact/dimension tables, relationships, business rules) and reusable transformation layers supporting self-serve analytics.
- Deployed a lightweight NLP classification model to auto-tag and route incoming donor/program inquiries, reducing manual triage effort by 25%.
- Built automated data quality checks that eliminated ~12 hrs/week of cleanup and reduced downstream report defects by an estimated 35%.

Texas A&M University—Corpus Christi

Corpus Christi, TX

Graduate Research Assistant — AI/ML Research Engineer / Jan 2024–May 2025

Supervisor: Prof. Dulal Kar, Ph.D. (Dept. Chair, CS) | CI 321 | (361) 825-5878 | dulal.kar@tamucc.edu | faculty.tamucc.edu/dkar

- Spearheaded an automation-first, taxonomy-driven LLM red-teaming framework executing 3,000+ adversarial interactions across six model/attack configurations, with database-backed logging for replayable analysis.
- Built a closed-loop red-teaming pipeline integrating GPT-4o-mini as a red-team agent, quantized Llama-3.2-1B-Instruct as the target model, and ensemble safety graders (LlamaGuard, DeBERTa reward, Detoxify, RoBERTa) into a composite harm scoring engine.
- Engineered three automated attack regimes (OpenAI-style baseline, multi-turn PYRIT Crescendo, jailbreak templates) using few-shot prompting and template orchestration.
- Reduced attack success rates by 20–50 percentage points through safety fine-tuning of the target model while avoiding excessive refusals.
- Architected an end-to-end MLOps pipeline for audio-based UAV detection: parallelized STFT/Mel spectrogram preprocessing for 50,000+ signals with Spark/Python, cutting feature extraction time from 15 hours to under 5 hours (67% faster).
- Delivered 97.2% F1-score and 90%+ accuracy for real-time unauthorized UAV detection across 8 drone types, with MLflow-driven experiment tracking and <100 ms inference.
- Built production-oriented RAG systems using LangChain, Hugging Face Transformers, and FAISS, achieving sub-200 ms query latency.
- Designed SQL data models and logging schemas supporting 3,000+ automated runs; query tuning cut research data prep time by 40%.
- Built Power BI operational dashboards (datasets + refresh patterns) for lab KPIs and throughput, reducing ad-hoc report requests by 30%.
- Published and presented peer-reviewed research on UAV acoustic detection (ICUAS 2025) and automated taxonomy-driven LLM red teaming.

Tata Consultancy Services (TCS)

Hyderabad, India

Data Engineer / Jul 2021–Jul 2023

Reference: M V P B Srinivas (Mentor, TCS) | +91 9550631225 | srinivas.mvpb@tcs.com

- Architected cloud-native ETL pipelines on AWS (S3, Redshift, Glue, Lambda) processing 15M+ legal and financial records per month, improving pipeline latency by 40% through optimized partitioning, parallel loads, and incremental processing.
- Led critical data migration and integration for a Fortune 500 client (Xerox), achieving 99.9% data availability and enabling near real-time analytics dashboards for 500+ enterprise users.
- Developed Python-based intelligent web scrapers (BeautifulSoup, Selenium) with NLP-powered content extraction, automating ingestion of 10K+ legal documents monthly and reducing manual document processing by 35%.
- Collaborated with ML teams to design feature engineering pipelines and data quality frameworks (schema validation, anomaly detection, automated profiling), increasing downstream ML model accuracy by 15%.
- Built executive-facing Power BI/Tableau dashboards with real-time KPI tracking, reducing document turnaround time by 35%.
- Engineered batch and near real-time ingestion frameworks with Airflow orchestration, reducing manual intervention and improving pipeline reliability.
- Optimized SQL/Redshift workloads and warehouse query patterns, improving high-volume analytics query execution time by ~25%.
- Designed multi-layer ELT (raw → curated → consumption) patterns transferable to Azure Synapse/Snowflake warehouse architectures.

KEY PROJECTS

Automated Taxonomy-Driven Red Teaming for LLMs

Jan 2025–Present

Python, GPT-4o-mini, Llama-3.2, PYRIT, LlamaGuard, FAISS

- Designed an enterprise-grade adversarial testing framework with persistent seed/prompt/response logging and taxonomy-aware grading for scalable LLM safety evaluation.
- Orchestrated 3,000+ adversarial interactions across six configurations using a GPT-4o-mini red-team agent against quantized Llama-3.2-1B-Instruct.
- Built a weighted ensemble safety scorer combining DeBERTa reward, LlamaGuard, Detoxify, and RoBERTa signals into a composite harm metric (L0–L3), demonstrating 20–50 percentage-point reductions in attack success rate via safety fine-tuning.

UAV Audio Detection using Deep Learning

Jun 2024–Jan 2025

PyTorch, Spark, MLflow, STFT/Mel, CNN/RNN/CRNN/VGG19

- Developed spectrogram-based CNN, RNN, CRNN, and VGG19 models using STFT and Mel spectrogram transformations, achieving 97.2% F1-score and 90%+ accuracy across 8 drone types (ICUAS 2025).
- Engineered scalable data augmentation and preprocessing pipelines with Spark/Python, improving model generalization by 12% and enabling <100 ms inference.

Supply Chain Analytics Dashboards

May 2025–Aug 2025

SQL Server, Python, Power BI

- Delivered a full-stack analytics solution integrating SQL Server, Python preprocessing, and automated Power BI dashboards for 200+ operational stakeholders; reduced manual reporting by ~30%.

Enterprise Data Ingestion & Warehouse Foundation

AWS/Azure-pattern, Python, SQL

- Designed multi-layer ELT (raw → curated → consumption) with parallel loads and incremental strategies, reducing batch overruns by an estimated 40%.
- Built dimensional models and reusable SQL transformation frameworks, cutting new-source onboarding time by ~30%.

API-Integrated Analytics Refresh Framework

Python, REST/JSON, SQL, Power BI

- Engineered REST/JSON API ingestion with schema alignment and retry logic feeding SQL models and Power BI datasets, improving freshness SLAs to under 24 hours.

Production RAG Knowledge Systems

LangChain, FAISS, Hugging Face

- Built production-oriented RAG pipelines with dense retrieval and vector indexing, achieving sub-200 ms query latency for enterprise-style document knowledge management.

EDUCATION

Texas A&M University–Corpus Christi — M.S., Computer Science

Aug 2023–May 2025

GPA: 3.93/4.0 | Leadership Scholarship Recipient | Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, AI Safety, Distributed Systems, Cloud Computing, Databases

Vasavi College of Engineering — B.E., Information Technology

Aug 2017–Aug 2021

GPA: 3.64/4.0

CERTIFICATIONS

- **Microsoft Certified: Power Platform Developer Associate** — Credential ID: 29F3695AA77BF974 | Feb 2026–Feb 2027
- **Microsoft Certified: Power BI Data Analyst Associate** — Certification ID: 8C3D63-F1C5AB | Jun 2025–Jul 2027
- **Microsoft Certified: Azure Administrator Associate** — Certification ID: 7052K5-971844 | Feb 2023–Feb 2027
- UAV Audio Detection and Identification Using STFT Spectrograms with Deep Learning, ICUAS 2025 (peer-reviewed).
- Automated Taxonomy-Driven Red Teaming for Large Language Models, under review/in preparation, 2025.